

Hopf bifurcation problem

$$\dot{z} = (\lambda + i\omega)z + (a + ib)|z|^2 z \quad \leftarrow \text{"normal form"}$$

$$z = x + iy = r e^{i\theta} \Rightarrow \begin{aligned} \dot{r} &= \lambda r + a r^3 \\ \dot{\theta} &= \omega + b r^2 \end{aligned}$$

 $a > 0 \Rightarrow$ subcritical $a < 0 \Rightarrow$ supercriticalNormal Form TransformationExample 1:

$$\dot{W} = i\omega W + A W^2$$

Let $W = Z + \alpha Z^2$, where we will choose α to "simplify" problem in Z -coordinates

$$\dot{W} = \dot{Z}(1 + 2\alpha Z) = i\omega(Z + \alpha Z^2) + A(Z + \alpha Z^2)^2$$

$$\Rightarrow \dot{Z} = \underbrace{(1 + 2\alpha Z)^{-1}} \left[i\omega(Z + \alpha Z^2) + A Z^2 + \mathcal{O}(Z^3) \right]$$

$$\frac{1}{1 + 2\alpha Z} = 1 - 2\alpha Z + \mathcal{O}(Z^2)$$

$$\dot{Z} = i\omega Z(1 - 2\alpha Z) + i\omega \alpha Z^2 + A Z^2 + \mathcal{O}(Z^3)$$

Lecture 13 p.2

$$\begin{aligned}\dot{Z} &= i\omega Z - 2\alpha i\omega Z^2 + i\omega\alpha Z^2 + AZ^2 + O(Z^3) \\ &= i\omega Z + (A - i\alpha\omega)Z^2 + O(Z^3)\end{aligned}$$

Choose α s.t. $A - i\alpha\omega = 0$

$$\Rightarrow \dot{Z} = i\omega Z + O(Z^3)$$

Turns out this strategy works for all quadratic terms.

Example 2:

$$\dot{Z} = i\omega Z + A|Z|^2 Z$$

↙ "normal form"

Can we remove this cubic term?

$$\text{Try } Z = Y + \underbrace{\alpha |Y|^2 Y}_{\substack{Y^2 \bar{Y} \\ Y^2 \bar{Y}}} ; \quad \bar{Z} = \bar{Y} + \bar{\alpha} |Y|^2 \bar{Y}$$

$$\begin{aligned}\dot{Z} &= \dot{Y}(1 + 2\alpha |Y|^2) + \dot{\bar{Y}}(\alpha Y^2) = i\omega(Y + \alpha |Y|^2 Y) \\ &\quad + A|Y|^2 Y + O(Y^4)\end{aligned}$$

$$\begin{aligned}\dot{\bar{Z}} &= \dot{\bar{Y}}(1 + 2\bar{\alpha} |Y|^2) + \dot{Y}(\bar{\alpha} \bar{Y}^2) = -i\omega(\bar{Y} + \bar{\alpha} |Y|^2 \bar{Y}) \\ &\quad + \bar{A}|Y|^2 \bar{Y} + O(Y^4)\end{aligned}$$

$$\begin{pmatrix} 1+2\alpha|Y|^2 & \alpha Y^2 \\ \bar{\alpha}\bar{Y}^2 & 1+2\bar{\alpha}|Y|^2 \end{pmatrix} \begin{pmatrix} \dot{Y} \\ \dot{\bar{Y}} \end{pmatrix} = \begin{pmatrix} i\omega Y + (i\omega\alpha + A)|Y|^2 Y + O(Y^4) \\ -i\omega\bar{Y} + (-i\omega\bar{\alpha} + \bar{A})|Y|^2 \bar{Y} + O(Y^4) \end{pmatrix}$$

$$(\mathbf{I} + \epsilon \mathbf{M})^{-1} = \mathbf{I} - \epsilon \mathbf{M} + O(\epsilon^2)$$

$$\begin{pmatrix} \dot{Y} \\ \dot{\bar{Y}} \end{pmatrix} = \begin{pmatrix} 1-2\alpha|Y|^2 & -\alpha Y^2 \\ -\bar{\alpha}\bar{Y}^2 & 1-2\bar{\alpha}|Y|^2 \end{pmatrix} \begin{pmatrix} i\omega Y + (i\omega\alpha + A)|Y|^2 Y + O(Y^4) \\ -i\omega\bar{Y} + (-i\omega\bar{\alpha} + \bar{A})|Y|^2 \bar{Y} + O(Y^4) \end{pmatrix}$$

$$\begin{aligned} \dot{Y} &= (1-2\alpha|Y|^2)i\omega Y + (i\omega\alpha + A)|Y|^2 Y + i\omega\alpha|Y|^2 Y + O(Y^4) \\ &= i\omega Y + \underbrace{(-2i\alpha\omega + i\omega\alpha + A + i\omega\alpha)}_{=A}|Y|^2 Y + O(Y^4) \end{aligned}$$

can't be helped!

$$\left\{ \begin{aligned} \dot{X} &= \lambda X - \omega Y + XY - X^2 \\ \dot{Y} &= \lambda Y + \omega X - Y^2 \end{aligned} \right\} \quad \text{Hopf at } \lambda=0$$

(see problem 8, 2, 12)

is it sub or super critical? Need to know $\text{Re}(A) = a$

$$\left. \begin{aligned} f &= XY - X^2 \\ g &= -Y^2 \end{aligned} \right\} \quad \text{Iba} = \frac{1}{\omega} \begin{bmatrix} f_{xy} & f_{xx} \end{bmatrix} = -\frac{2}{\omega}$$

$\swarrow \quad \searrow$
 $1 \quad -2$

sub. if $\omega < 0$
 super. if $\omega > 0$

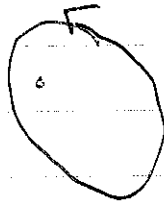
Lecture 13 p.4

$$\left. \begin{aligned} \dot{x} &= \lambda x - y + xy - x^2 \\ \dot{y} &= \lambda y + x - y^2 \end{aligned} \right\} \quad \omega = +1 \Rightarrow \text{supercritical}$$

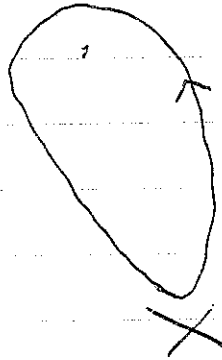
$$\lambda = 0$$

center at $(x, y) = (0, 0)$ ~~general~~
saddle at $(x, y) \approx (0, 6, -0, 75)$

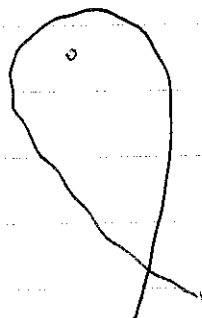
$$\lambda = 0,01$$



$$\lambda = 0,02$$



$$\lambda = 0,026 \dots$$



limit cycle
destroyed
in homoclinic
bifurcation

$$\dot{x} = \lambda x - \omega y + xy - x^2$$

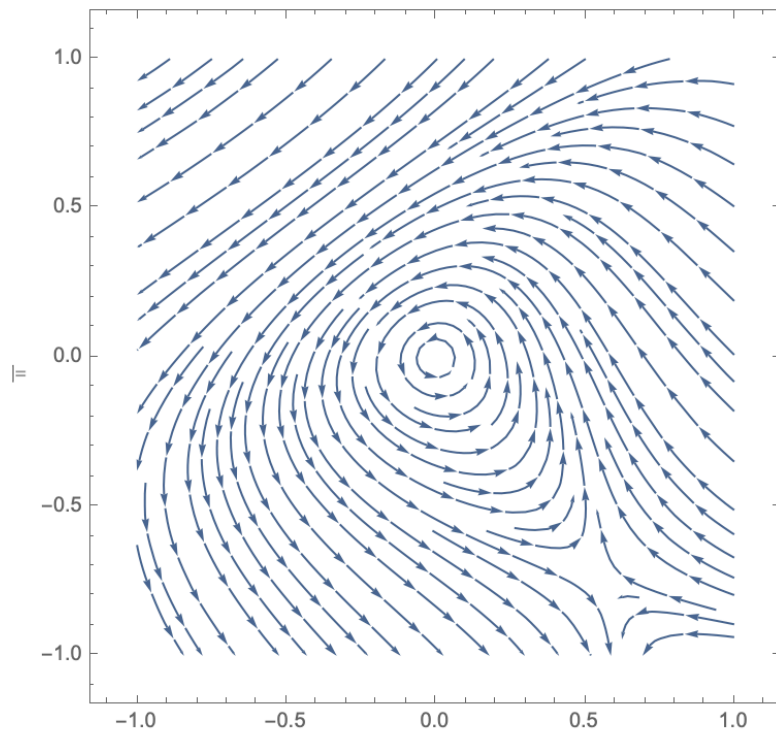
$$\dot{y} = \lambda y + \omega x - y^2$$

**supercritical Hopf bifurcation for
omega>0; here we set omega=1**

:=

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omega = 1;
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```
st = StreamPlot[{lambda * x - omega * y + x * y - x ^ 2, lambda * y + omega * x - y ^ 2} /. lambda -> 0.0, {x, -1, 1}, {y, -1, 1}]
```

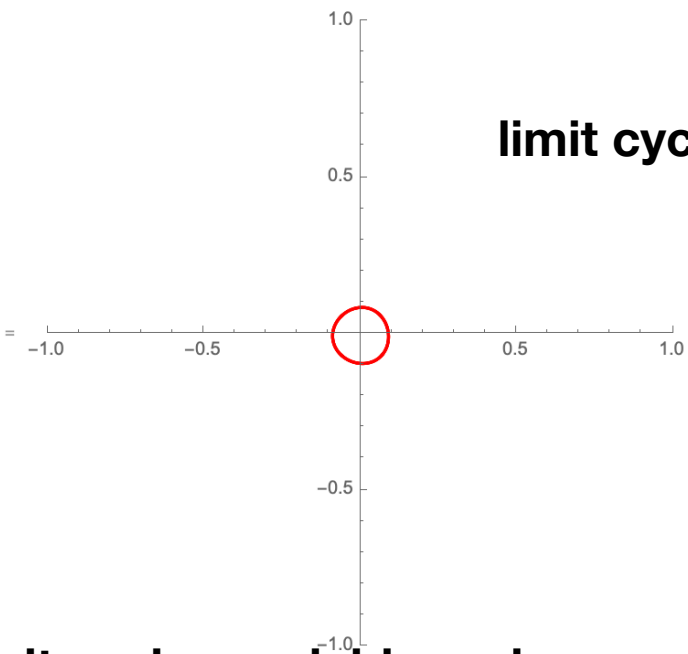


phase portrait for lambda=0

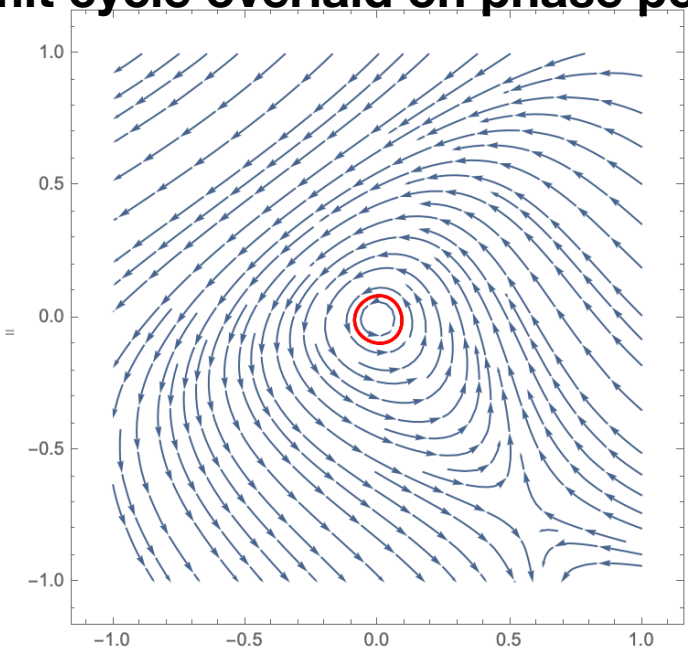
```
s = NDSolve[{x'[t] == lambda * x[t] - omega * y[t] + x[t] * y[t] - x[t]^2, y'[t] == lambda * y[t] + omega * x[t] - y[t]^2, x[0] == y[0] == 0.25} /. lambda -> 0.001, {x, y}, {t, 5000}];
p = ParametricPlot[Evaluate[{x[t], y[t]} /. s], {t, 5000 - 10 * Pi, 5000}, PlotRange -> {{-1, 1}, {-1, 1}}, PlotStyle -> Red]
Show[{st, p}]
```

$\lambda = 0.001$

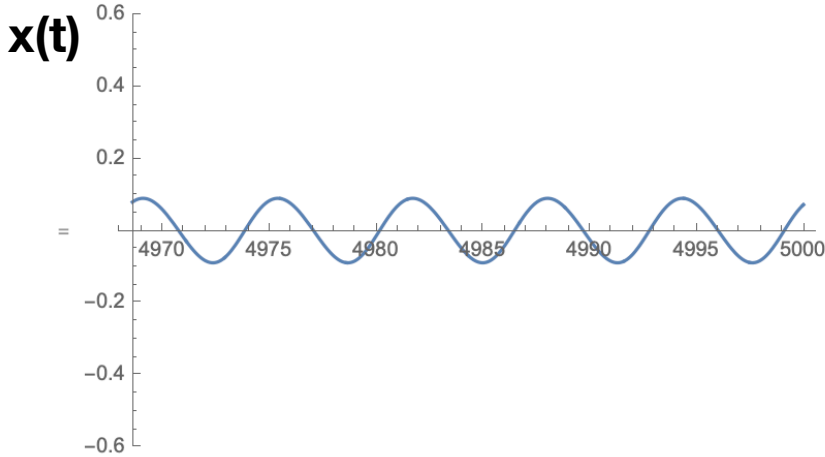
limit cycle in phase plane



limit cycle overlaid on phase portrait

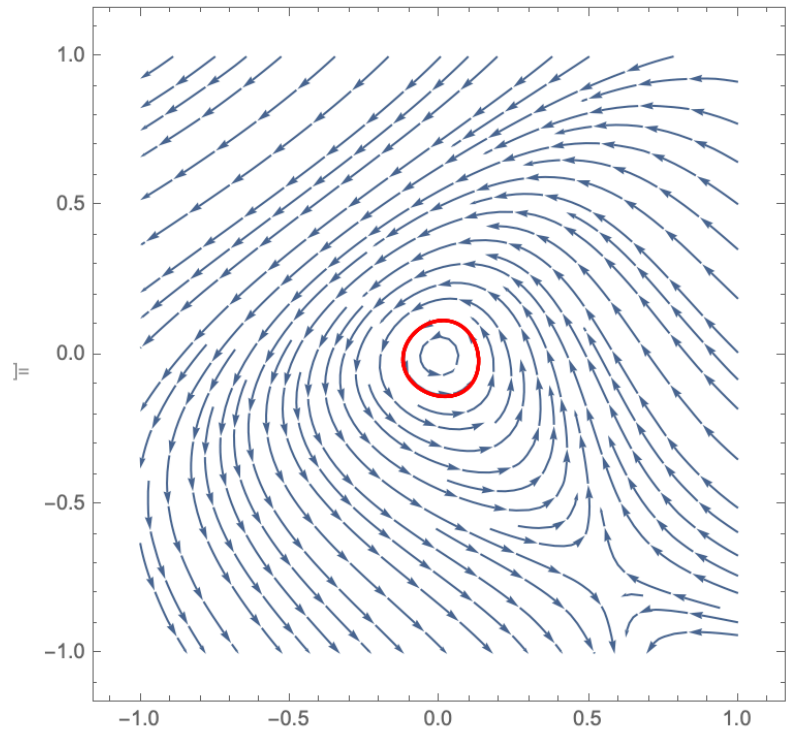


```
Plot[x[t] /. s, {t, 5000 - 10 * Pi, 5000}, PlotRange -> {-0.6, 0.6}]
```

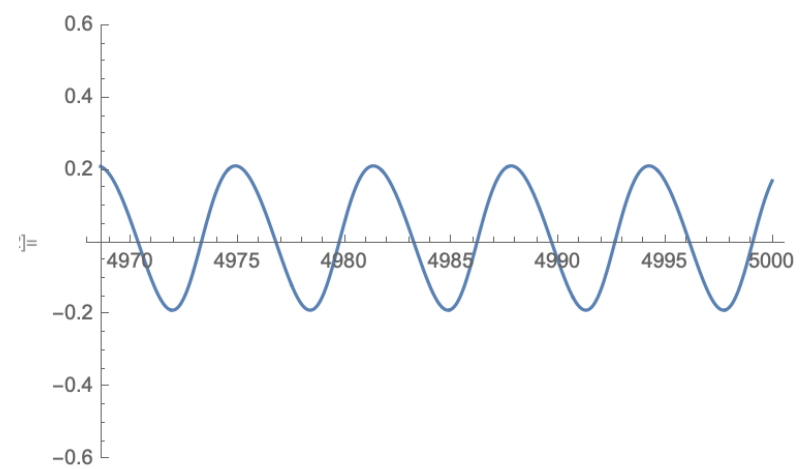
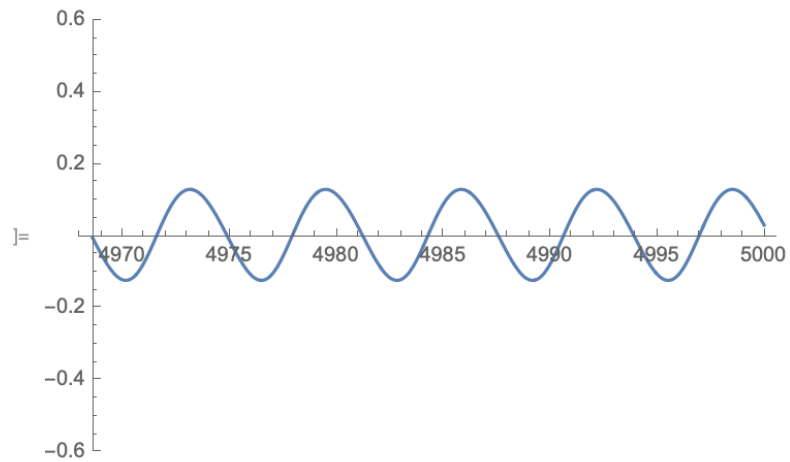
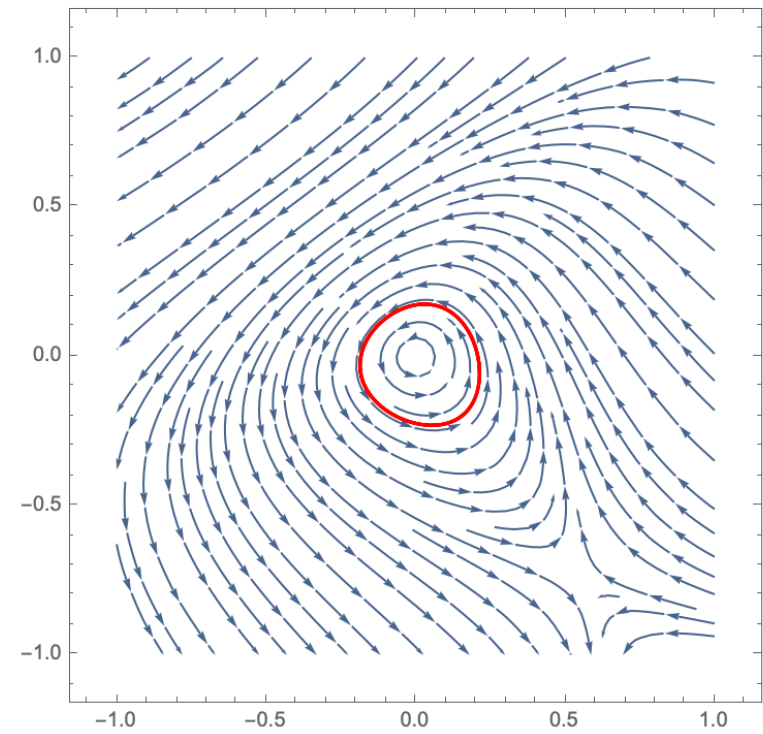


$$\lambda = 0.002$$

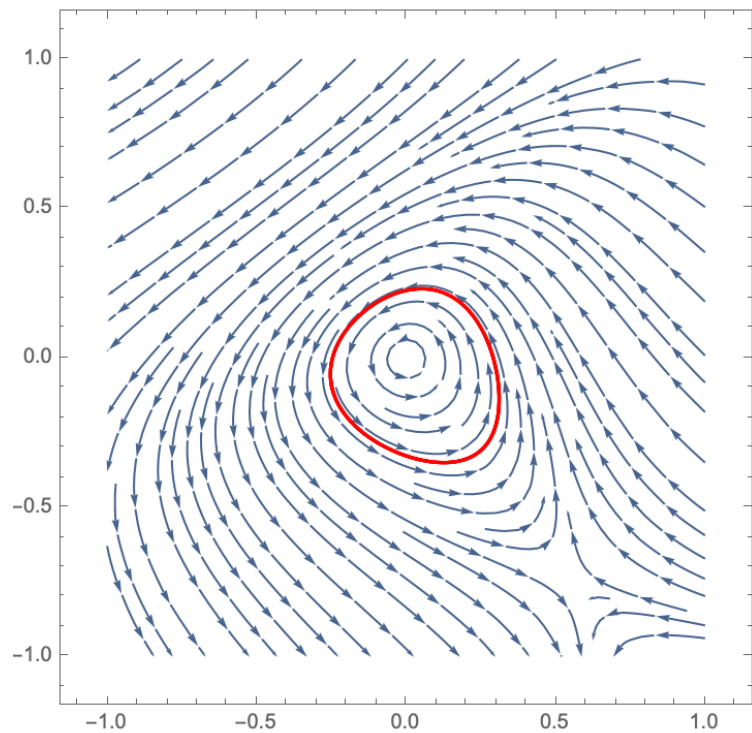
$-1.0 \leq$



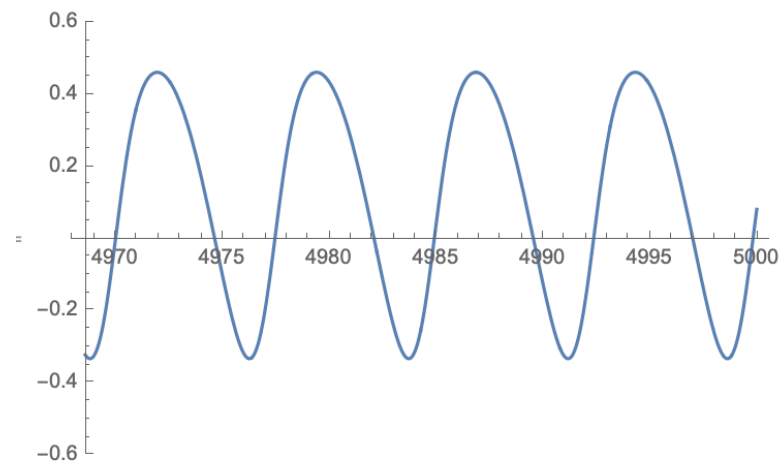
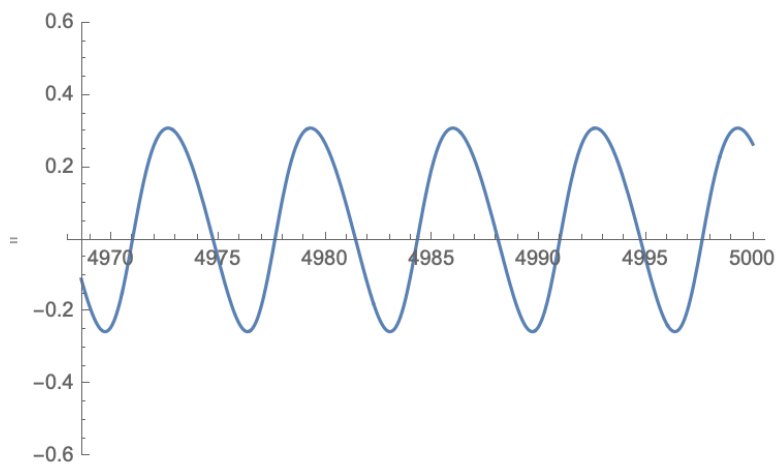
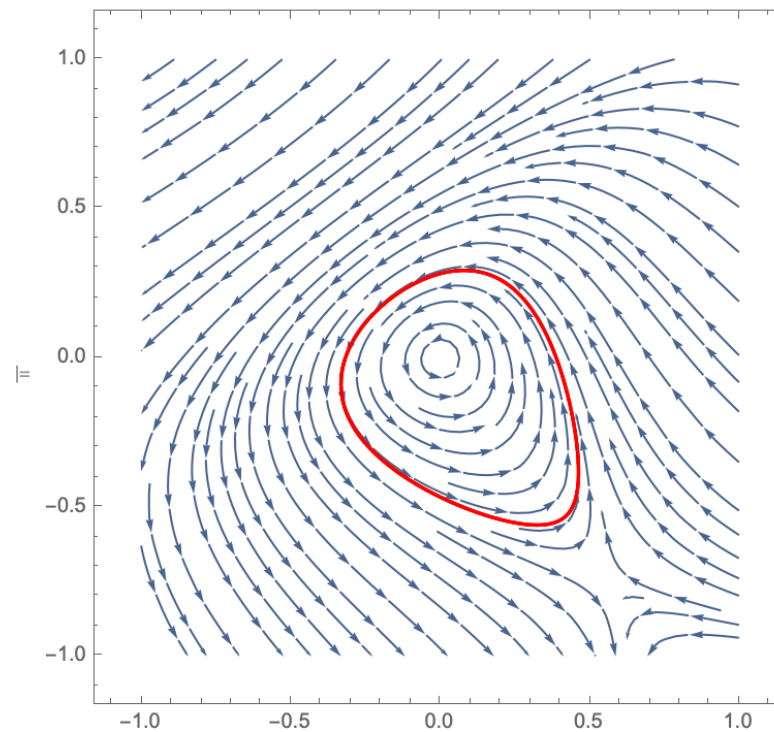
$$\lambda = 0.005$$



$$\lambda = 0.01$$

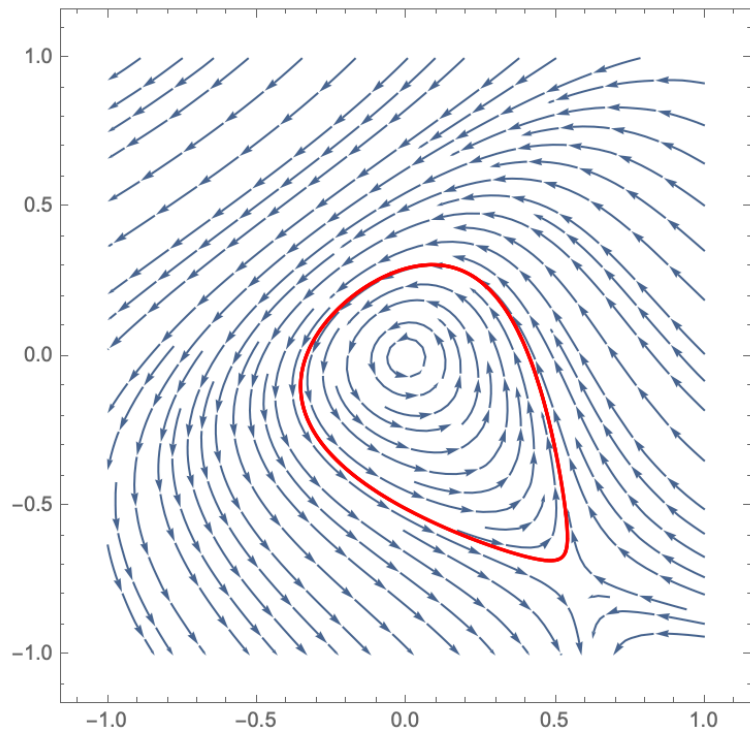


$$\lambda = 0.02$$

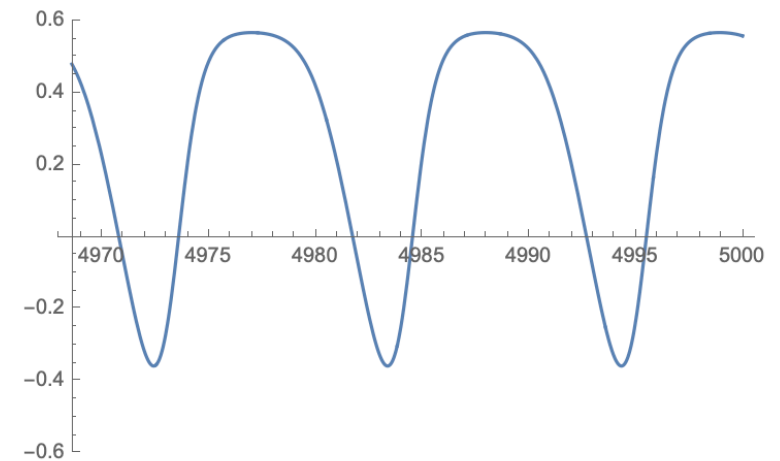
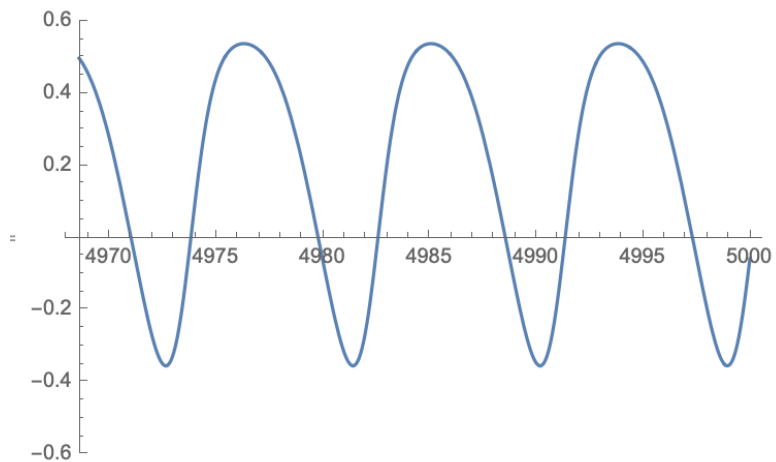
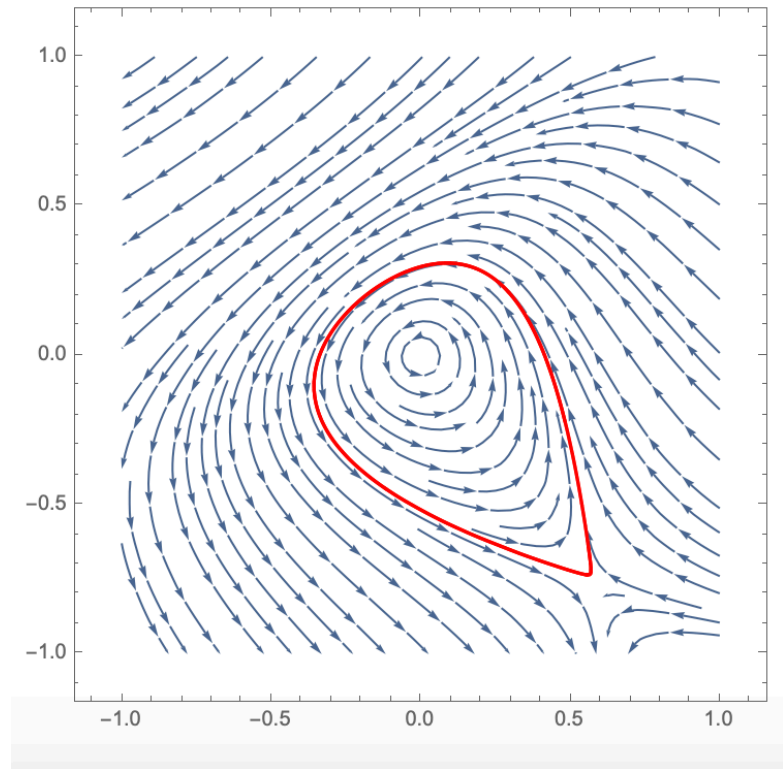


limit cycle is approaching its destruction....

$$\lambda = 0.025$$



$$\lambda = 0.0262$$



Lorenz eqns (Lorenz '63)

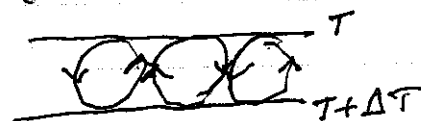
$$\left. \begin{aligned} \dot{x} &= \sigma(y-x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz \end{aligned} \right\}$$

$\sigma, r, b > 0$
 "standard parameters"
 $\sigma = 10, b = 8/3$

$r = \text{bifurcation parameter}$

- only 2 nonlinear terms
- symmetry $(x, y, z) \rightarrow (-x, -y, z)$

Eqns. originally came from a model of Rayleigh-Bénard convection, where



$\sigma = \text{Prandtl number}$

$r \propto \text{Rayleigh number}$ (how much fluid is heated)

b depends on aspect ratio of convective rolls

x is related to amplitude of convection
 y & z are related to temperature

specifically:

$$\psi \sim X \sin\left(\pi a \frac{x}{H}\right) \sin\left(\pi \frac{z}{H}\right)$$

stream function
 $u = \frac{\partial \psi}{\partial z}, w = -\frac{\partial \psi}{\partial x}$

$$\theta \sim Y \cos\left(\pi a \frac{x}{H}\right) \sin\left(\pi \frac{z}{H}\right) \rightarrow \frac{Z}{\sqrt{2}} \sin\left(2\pi \frac{z}{H}\right)$$

temp. deviation from linear conduction profile